

LGM: A Calibrated, Certifiable Factorized Neural Hawkes Process for Limit-Order-Book Simulation

Anonymous submission

Abstract

Neural marked temporal point processes (MTPPs) predict the next limit-order-book (LOB) event well but *simulate* order flow poorly: in free roll-out they over-disperse or explode, and their stability cannot be honestly certified. We introduce LGM (Linear Ground $\times$ soft-max Marks), which factorizes the per-type intensity into a *linear* scalar ground rate and a *deep* soft-max mark head. Because the marks live on the probability simplex they are rate-neutral, so the total rate remains a pure linear Hawkes process regardless of mark-net nonlinearity. This preserves the exact stationary-mean formula, which we *pin* analytically to solve the train/simulate calibration mismatch, and yields a gauge-free branching ratio as a closed-form stability certificate. On Gemini ETH-USD data (62 event types), LGM is simultaneously calibrated (free-roll rate within $\sim 7\%$ of real), a competitive predictor, clustered (correct Fano profile), and certifiable—the first model in our study to achieve all of these together.

Introduction

Order-book simulation underpins market-making research, backtesting, and the reinforcement learning of execution policies (Avellaneda and Stoikov 2008). Self- and mutually-exciting Hawkes processes are the classical model of clustered order flow (Hawkes 1971; Bacry, Mastromatteo, and Muzy 2015), and neural MTPPs extend them with expressive state (Mei and Eisner 2017). Yet a model that predicts the next event accurately can still fail as a *simulator*: under free roll-out the event rate runs away or collapses, and the stability of the recurrent dynamics cannot be read off the trained weights. We trace this to two causes—a train/simulate calibration mismatch and a gauge pathology in the rate read-out—and remove both by construction.

The LGM Factorization

The per-type intensity factorizes into a timing rate and a mark distribution,

$$\lambda_k(t) = \Lambda(t) p(k | t), \quad \sum_k \lambda_k(t) = \Lambda(t), \quad (1)$$

where the ground rate is a multi-timescale *linear* Hawkes process and the marks are a deep soft-max,

$$\Lambda(t) = \mu_0 + \sum_m a_m s_m(t), \quad p(\cdot | t) = \text{softmax}(z_\theta(t)). \quad (2)$$

LGM: $\lambda_k(t) = \Lambda(t) \cdot p(k | t)$

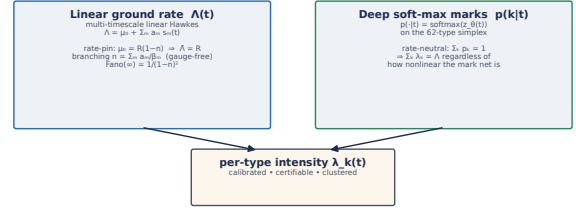


Figure 1: LGM factorizes the per-type intensity into a linear, rate-pinned ground rate $\Lambda(t)$ and a deep, rate-neutral soft-max over the 62 event types $p(k | t)$. Nonlinearity is confined to the simplex, where it cannot perturb the calibrated rate or the branching-ratio certificate.

Since the soft-max lies on the simplex, $\sum_k p_k = 1$, the marks are *rate-neutral*: the total rate Λ is unaffected by the mark-net nonlinearity. The stationary mean therefore survives in closed form,

$$\bar{\Lambda} = \frac{\mu_0}{1-n}, \quad n = \sum_m \frac{a_m}{\beta_m}, \quad (3)$$

and we *pin* it by setting $\mu_0 = R_{\text{target}}(1-n)$, so $\bar{\Lambda} = R_{\text{target}}$ by construction. The branching ratio n is gauge-free and serves as a stability certificate, with $\text{Fano}(\infty) = 1/(1-n)^2$.

Results

Table 1 reports the headline comparison on Gemini ETH-USD (62 event types, real rate ≈ 2.38 ev/s). LGM is uniquely calibrated *and* a competitive predictor *and* clustered *and* certifiable; the branching ratio is a single interpretable knob that sets the Fano-vs-scale curve via $1/(1-\rho)^2$.

Discussion

LGM removes the two failure modes of neural LOB simulators by construction: the analytic rate-pin (Eq. 3) fixes the train/simulate calibration mismatch without a stateful loader, and the direct linear ground read-out keeps the branching ratio gauge-free and measurable from weights. Honest caveats remain: return tails are $\sim 2\times$ lighter than the robust empirical target, the model is mildly over-reflexive,

Table 1: Headline comparison on Gemini ETH–USD (62 event types; real rate ≈ 2.38 ev/s). GenAcc: genuine next-event accuracy.

Model	GenAcc	Fano(1s)	ρ	roll rate	sim status
Compound Hawkes	0.18	8.8	0.64	~ 2	stable, no mem.
s2p2 (predictor)	0.32	41	—	—	over-disperses
NMH / GMH	0.26	expl.	1300/0.8	52–100	explode / Pois.
LGM ($n=0.86$)	0.29	6.0	0.86	2.22	calibrated

and raw 1 s kurtosis/skew are outlier-dominated and should be read winsorized or at ≥ 5 s buckets. Action-conditioning toward a market-making world model is the natural next step.

References

- Avellaneda, M.; and Stoikov, S. 2008. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3): 217–224.
- Bacry, E.; Mastromatteo, I.; and Muzy, J.-F. 2015. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(01).
- Hawkes, A. G. 1971. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1): 83–90.
- Mei, H.; and Eisner, J. 2017. The neural Hawkes process: A neurally self-modulating multivariate point process. In *Advances in Neural Information Processing Systems (NeurIPS)*.